

Experiment no-20

20. Implement a basic information retrieval system using TF-IDF (Term Frequency-Inverse Document Frequency) for document ranking using python.

PROGRAM:

```
[6]
✓ 22s
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Sample documents
documents = [
    "Natural Language Processing is a branch of Artificial Intelligence.",
    "Machine Learning helps computers learn from data.",
    "Python is widely used for NLP and Machine Learning."
]

# User query
query = input("Enter your search query: ")

# TF-IDF Vectorization
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(documents)

# Transform query
query_vector = vectorizer.transform([query])

# Compute similarity
similarity = cosine_similarity(query_vector, tfidf_matrix)

# Display ranked documents
scores = similarity.flatten()

print("\nRanked Documents:")
for i in scores.argsort()[::-1]:
    print(f"Document {i+1}: {documents[i]}")
    print(f"Similarity Score: {scores[i]:.4f}\n")
```

OUTPUT:

```
... Enter your search query: natural language processing

Ranked Documents:
Document 1: Natural Language Processing is a branch of Artificial Intelligence.
Similarity Score: 0.6292

Document 3: Python is widely used for NLP and Machine Learning.
Similarity Score: 0.0000

Document 2: Machine Learning helps computers learn from data.
Similarity Score: 0.0000
```